

EXCLUSION LAWS FOR CONSECUTIVE ARTIN PRIMES IN ARBITRARY BASES

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ABSTRACT. Say that a prime p is *Artin base a* if a is a primitive root modulo p . We prove that for every non-square base a there is an explicit set of gap classes at which two primes can *never* both be Artin base a : if the gap $g = q - p$ lies in one of these classes, the quadratic character χ_a satisfies $\chi_a(q) = -\chi_a(p)$, so one of the two primes is a quadratic residue base a and hence not a primitive root. We classify these *exclusion classes* completely in terms of the factorization of the discriminant of $\mathbb{Q}(\sqrt{a})$ into prime discriminants, and we deduce a clean dichotomy: base a admits an exclusion law if and only if the squarefree part d of a satisfies $d \equiv 0 \pmod{2}$, $d \equiv 3 \pmod{4}$, or $3 \mid d$. In particular bases with $d \equiv 1 \pmod{4}$ odd and $3 \nmid d$ — the smallest being $d = 5, 13, 17, 29$ — admit *no* exclusion law at all. The classification is verified by exhaustive computation for all non-square $2 \leq a \leq 30$, and the exclusion law itself is verified on 63,105,745 consecutive prime pairs below 10^9 across seven bases, with zero exceptions.

We then measure, for eleven bases, the consecutive-pair Artin correlation $\delta(a)$ over all 50,847,531 consecutive prime pairs below 10^9 . Here the outcome is negative in an instructive way. One might expect bases with denser exclusion classes to exhibit stronger anticorrelation; they do not. The association with exclusion density is weak ($r = 0.23$), and base 5, which admits no exclusion law whatsoever, has the *largest* $|\delta|$ of the eleven. Instead $|\delta(a)|$ is governed by the size of the conductor f of χ_a , with $r(\log f, |\delta|) = -0.96$. The data strongly indicate that the two phenomena are distinct: the exclusion law is locally absolute but globally negligible, while the global correlation is a conductor-size effect inherited from the Lemke Oliver–Soundararajan bias.

1. INTRODUCTION

Artin’s conjecture on primitive roots asserts that for a non-square integer $a \neq -1$ the set of primes p for which a is a primitive root modulo p has a positive density, given by an explicit rational multiple of Artin’s constant. It is known under GRH by Hooley [8], and unconditionally for at least one of any three multiplicatively independent bases by Gupta–Murty [5] and Heath-Brown [7]; see Moree’s survey [12] for the extensive literature, and [2, 13] for recent uniform versions.

Write

$$A_a = \{p \text{ prime} : a \text{ is a primitive root mod } p\},$$

and call $p \in A_a$ an *Artin prime base a* . All results above concern a *single* prime at a time. The behaviour of Artin primes in *pairs* has been studied much less. Garcia, Kahoro and Luca [3] (see also [4]) investigated primitive-root bias for twin primes; Pollack [14] and Baker–Pollack [1] showed, under GRH, that there are infinitely many bounded-gap runs of primes sharing a primitive root.

Date: August 17, 2026.

2020 Mathematics Subject Classification. Primary 11A07; Secondary 11N05, 11N13, 11Y16, 11Y60.

Key words and phrases. Artin’s conjecture, primitive roots, consecutive primes, quadratic characters, prime discriminants, Lemke Oliver–Soundararajan bias.

The starting point of this paper is an elementary but apparently unrecorded observation. Since a quadratic residue cannot generate the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$ of even order $p - 1$, membership in A_a forces $\chi_a(p) = -1$, where χ_a is the quadratic character attached to $\mathbb{Q}(\sqrt{a})$. Because χ_a is a character modulo the conductor f of that field, the value $\chi_a(p)$ depends only on $p \bmod f$. Consequently, if a shift g has the property that it *reverses* χ_a on every admissible residue class modulo f , then for any two primes p and $q = p + g'$ with $g' \equiv g \pmod{f}$ we get $\chi_a(q) = -\chi_a(p)$, so at least one of $\chi_a(p), \chi_a(q)$ equals $+1$ — and that prime is not Artin base a . Two primes at such a gap can never *both* be Artin base a . We call this an *exclusion law* (Theorem 5).

Since the observation is elementary, we have searched the literature for it specifically before claiming novelty. Moree’s hundred-page survey [12] uses the quadratic obstruction $\chi_a(p) = -1$ throughout — it is the source of the entanglement correction in Hooley’s density — but always as a condition on a *single* prime; shifts of the argument of χ_a , and the resulting constraints on *pairs* of primes, do not appear. Garcia–Kahoro–Luca [3] and its sequel [4] concern the single gap $g = 2$ and a different statistic (a bias in $\varphi(p - 1)/(p - 1)$ between twin-prime components), not the joint Artin status of the pair; the reversing mechanism plays no role there. Pollack [14] and Baker–Pollack [1] produce bounded-gap runs of primes sharing a primitive root, but impose no structure on the gap classes and do not identify forbidden ones. We likewise searched for forbidden-gap statements for quadratic characters in other contexts — consecutive quadratic residues and non-residues have a classical literature — and found constraints on character values along progressions, but no statement coupling a gap *class* to the joint primitive-root status of a pair of primes, which is the content here. In that sense “apparently unrecorded” above is meant as the outcome of a search, not as an impression.

Our main theorem determines, for every base, exactly which gap classes are exclusion classes.

Theorem 1 (Classification; see Theorem 8). *Let a be a non-square positive integer with squarefree part d , let D be the discriminant of $\mathbb{Q}(\sqrt{d})$, and factor $D = D_1 D_2 \cdots D_k$ into prime discriminants. Then an even shift g is an exclusion class for base a if and only if no component D_i is indeterminate at g and an odd number of the components reverse at g , where the behaviour of each component is given by Table 1.*

The classification is sharp enough to answer the qualitative question completely.

Corollary 2 (Dichotomy; see Corollary 9). *Base a admits at least one exclusion class if and only if d is even, or $d \equiv 3 \pmod{4}$, or $3 \mid d$. Equivalently, if and only if the prime-discriminant factorization of D contains one of $-4, \pm 8, -3$.*

Thus for $d \in \{5, 13, 17, 29, 37, 41, 53, 61, \dots\}$ there is no exclusion law at all, while for $d = 3$ (and $d = 12, 27, \dots$) fully half of the even gap classes modulo 12 are exclusion classes. This is a genuine structural dichotomy, not an artifact of small conductors, and its source is a degeneration of a Jacobi-sum obstruction at the single prime 3 (Remark 7).

In the second half of the paper we turn from the deterministic statement to a statistical one. For a base a define the consecutive-pair correlation

$$\delta(a) = P(p_{n+1} \in A_a \mid p_n \in A_a) - P(p_{n+1} \in A_a \mid p_n \notin A_a), \quad (1)$$

estimated over consecutive primes $p_n < p_{n+1} \leq x$. The bias of Lemke Oliver and Soundararajan [11] in the joint distribution of $(p_n \bmod m, p_{n+1} \bmod m)$ propagates, through the factorization of $p - 1$, into Artin status, so one expects $\delta(a) \neq 0$; and indeed we find $\delta(a) < 0$ for all eleven bases examined, with $|z| \geq 96$ throughout.

The natural guess is that the exclusion law drives this anticorrelation: bases whose exclusion classes capture a larger share of consecutive pairs should display a more negative δ . *This*

guess is false, and we regard its refutation as one of the contributions of this paper. Over all 50,847,531 consecutive prime pairs below 10^9 we find

- base 5, which admits *no* exclusion law, has the largest $|\delta|$ of the eleven bases;
- base 3, whose exclusion classes capture 53% of all consecutive pairs by weight, ranks only third;
- the mean $|\delta|$ over bases *with* an exclusion law is 0.0296, versus 0.0411 over bases *without* one — a ratio of 0.72, in the direction opposite to the guess;
- $r(\text{exclusion weight}, |\delta|) = 0.23$, whereas $r(\log f, |\delta|) = -0.96$.

The explanation is a matter of scale rather than of mechanism. An exclusion class imposes an absolute constraint, but only on a restricted set of gap classes; averaged over all pairs, the constraint is diluted. What survives in the average is the number of residue channels available to χ_a , which is controlled by f : the smaller the conductor, the fewer the classes over which the Lemke Oliver–Soundararajan transition bias is distributed, and the larger the resulting per-base correlation. We state this as a measured empirical law (Observation 12) rather than a conjectured exact power of f , because the data do not support a clean exponent.

Summary of results. Section 2 proves the exclusion law and the classification and establishes the dichotomy. Section 3 describes the computation. Section 4 reports the verification of the exclusion law at 10^9 . Section 5 reports the measurement of $\delta(a)$, the refutation of the exclusion-density hypothesis, and the conductor law. Section 6 discusses what we take the consequences to be.

2. THE EXCLUSION LAW AND ITS CLASSIFICATION

Throughout, $a \geq 2$ is an integer that is not a perfect square, $d = \text{sqf}(a)$ denotes its squarefree part, and

$$D = \begin{cases} d, & d \equiv 1 \pmod{4}, \\ 4d, & \text{otherwise,} \end{cases} \quad f = |D|, \quad (2)$$

so that D is the discriminant of $K = \mathbb{Q}(\sqrt{a}) = \mathbb{Q}(\sqrt{d})$ and f is the conductor of the associated quadratic character $\chi_a = \left(\frac{d}{\cdot}\right)$. By quadratic reciprocity χ_a is a Dirichlet character modulo f ; in particular

$$\chi_a(p) \text{ depends only on } p \bmod f \quad (p \nmid 2d). \quad (3)$$

We recall the elementary obstruction, which is the only number-theoretic input to Theorem 5.

Lemma 3. *Let $p > 2$ be a prime with $p \nmid a$. If a is a primitive root modulo p then $\chi_a(p) = -1$.*

Proof. If $\chi_a(p) = +1$ then $a \equiv b^2 \pmod{p}$ for some b , whence $a^{(p-1)/2} \equiv b^{p-1} \equiv 1 \pmod{p}$. Since $p > 2$, $(p-1)/2 < p-1$, so the multiplicative order of a is a proper divisor of $p-1$ and a is not a primitive root. $\square$

Definition 4. Let $R_f \subseteq (\mathbb{Z}/f\mathbb{Z})^\times$ be the set of residues attained by primes $p \nmid 2d$. For an even integer g , say that g is

- *reversing* for a if $\chi_a(r+g) = -\chi_a(r)$ for all $r \in R_f$ with $r+g \in R_f$;
- *preserving* for a if $\chi_a(r+g) = +\chi_a(r)$ for all such r ;
- *indeterminate* otherwise.

A reversing class $g \bmod f$ is called an *exclusion class* for base a .

Theorem 5 (Exclusion law). *Let g be reversing for a . Then there is no pair of primes $p, q > 2$ with $p, q \nmid a$, $q - p \equiv g \pmod{f}$, such that a is a primitive root modulo both p and q .*

TABLE 1. Behaviour of each prime-discriminant component of χ_a under the shift $r \mapsto r + g$. Here $q \geq 5$ is an odd prime.

component D_i	conductor	reversing iff	preserving iff
-4	4	$g \equiv 2 \pmod{4}$	$g \equiv 0 \pmod{4}$
8 or -8	8	$g \equiv 4 \pmod{8}$	$g \equiv 0 \pmod{8}$
-3	3	$g \not\equiv 0 \pmod{3}$	$g \equiv 0 \pmod{3}$
$q^*, q \geq 5$	q	never	$g \equiv 0 \pmod{q}$

Proof. By (3) and Definition 4, $\chi_a(q) = \chi_a(p + (q - p)) = -\chi_a(p)$. Hence exactly one of $\chi_a(p), \chi_a(q)$ equals +1, and by Lemma 3 the corresponding prime is not Artin base a . $\square$

Theorem 5 is unconditional and its proof is three lines; the content lies in determining the reversing classes, which we now do. Recall that every discriminant D factors uniquely as a product of *prime discriminants*

$$D = D_1 D_2 \cdots D_k, \quad D_i \in \{-4\} \cup \{8, -8\} \cup \{q^* : q \text{ odd prime}\}, \quad (4)$$

where $q^* = q$ if $q \equiv 1 \pmod{4}$ and $q^* = -q$ if $q \equiv 3 \pmod{4}$; the odd D_i are precisely the q^* for the odd primes $q \mid d$, and a factor $\pm 4, \pm 8$ occurs according to $d \pmod{8}$. Correspondingly $\chi_a = \prod_i \chi_{D_i}$, with χ_{D_i} of conductor $|D_i|$.

Proposition 6. *Table 1 is correct: each component behaves as stated, and in the cases not listed as reversing or preserving it is indeterminate. In particular a component of conductor 8 is indeterminate for $g \equiv 2, 6 \pmod{8}$, and a component q^* with $q \geq 5$ is indeterminate for every $g \not\equiv 0 \pmod{q}$.*

Proof. Conductor 4. $\chi_{-4}(r) = (-1)^{(r-1)/2}$ depends only on $r \pmod{4}$, and on $\{1, 3\}$ it is injective. A shift by $g \equiv 2 \pmod{4}$ interchanges 1 and 3, hence reverses; $g \equiv 0 \pmod{4}$ fixes both.

Conductor 8. $\chi_8(r) = +1$ iff $r \equiv \pm 1 \pmod{8}$, so the fibres are $\{1, 7\}$ and $\{3, 5\}$; the map $r \mapsto r + 4$ interchanges these two sets, hence reverses, while $r \mapsto r$ preserves. For $g \equiv 2 \pmod{8}$ we have $1 \mapsto 3$ (a reversal) but $3 \mapsto 5$ (a preservation), so the behaviour depends on the class and g is indeterminate; similarly for $g \equiv 6$. The same computation applies to χ_{-8} , whose +1-fibre is $\{1, 3\}$ and whose -1-fibre is $\{5, 7\}$, again interchanged by +4.

Conductor 3. $\chi_{-3}(r) = +1$ iff $r \equiv 1 \pmod{3}$. On the two admissible classes $\{1, 2\}$ the character is injective, so any $g \not\equiv 0 \pmod{3}$ moves $1 \leftrightarrow 2$ and reverses, while $g \equiv 0$ preserves.

Conductor $q \geq 5$. Suppose, for a contradiction, that $\chi_{q^*}(r+g) = \varepsilon \chi_{q^*}(r)$ for a fixed $\varepsilon \in \{\pm 1\}$, all r with $r, r+g \not\equiv 0 \pmod{q}$, and some $g \not\equiv 0 \pmod{q}$. Multiplying by $\chi_{q^*}(r)$ and summing over $r \pmod{q}$ gives

$$S(g) := \sum_{r \pmod{q}} \chi_{q^*}(r) \chi_{q^*}(r+g) = \varepsilon \cdot \#\{r : r, r+g \not\equiv 0\} = \varepsilon(q-2). \quad (5)$$

On the other hand, for $g \not\equiv 0 \pmod{q}$ the substitution $r = gt$ (legitimate since g is invertible) and the complete multiplicativity of χ_{q^*} give

$$S(g) = \sum_{t \pmod{q}} \chi_{q^*}(gt) \chi_{q^*}(g(t+1)) = \chi_{q^*}(g)^2 \sum_t \chi_{q^*}(t) \chi_{q^*}(t+1) = \sum_{t \pmod{q}} \chi_{q^*}\left(\frac{t+1}{t}\right),$$

the last sum being over $t \neq 0$. As t runs over the nonzero classes, $u = (t+1)/t = 1 + t^{-1}$ runs over all classes except $u = 1$, whence

$$S(g) = \sum_{u \neq 1} \chi_{q^*}(u) = \left(\sum_{u \bmod q} \chi_{q^*}(u) \right) - \chi_{q^*}(1) = 0 - 1 = -1.$$

Comparing with (5) forces $|-1| = q - 2$, i.e. $q = 3$. For $q \geq 5$ this is a contradiction, so no such ε exists and g is indeterminate. Finally $g \equiv 0 \pmod{q}$ clearly preserves. $\square$

Remark 7. The proof isolates precisely why the prime 3 behaves differently from all larger primes, and it is worth emphasising that this is structural rather than an artifact. The Jacobi-type sum $S(g)$ equals -1 for *every* q and every $g \not\equiv 0$, including $q = 3$; what fails at $q = 3$ is the *counting* side of (5), since $q - 2 = 1$ and $\varepsilon(q - 2) = \pm 1$ is then compatible with $S(g) = -1$. Concretely, only one residue class survives the requirement that r and $r + g$ both be prime to 3, and a sign condition imposed on a single class is vacuously uniform. Thus the reversing behaviour of the component -3 — which is what makes base 3 the most heavily constrained base — sits exactly at the boundary where the obstruction of Proposition 6 degenerates.

Theorem 8 (Classification of exclusion classes). *Let a be non-square with $D = D_1 \cdots D_k$ as in (4), and let g be even. Then g is reversing for a if and only if no D_i is indeterminate at g and the number of i with D_i reversing at g is odd; and g is preserving if and only if no D_i is indeterminate at g and that number is even.*

Proof. Since $\chi_a = \prod_i \chi_{D_i}$ and the moduli $|D_i|$ are pairwise coprime, the Chinese Remainder Theorem identifies R_f with the product of the component residue sets. If every component is reversing or preserving at g , then for every r , $\chi_a(r + g) = \prod_i \chi_{D_i}(r + g) = (\prod_i \varepsilon_i) \chi_a(r)$ where $\varepsilon_i = -1$ exactly for the reversing components; so χ_a is reversed iff the count of such i is odd. Conversely, suppose some component D_j is indeterminate at g , so there are r_j, r'_j in its residue set with $\chi_{D_j}(r_j + g) = -\chi_{D_j}(r_j)$ and $\chi_{D_j}(r'_j + g) = +\chi_{D_j}(r'_j)$. Fixing any admissible choices at the other components and varying only the j -th coordinate via CRT produces two residues $r, r' \in R_f$ at which χ_a is respectively reversed and preserved. Hence g is neither reversing nor preserving. $\square$

Corollary 9 (Dichotomy). *Base a admits at least one exclusion class if and only if*

$$d \equiv 0 \pmod{2}, \quad \text{or} \quad d \equiv 3 \pmod{4}, \quad \text{or} \quad 3 \mid d,$$

equivalently if and only if $\{-4, 8, -8, -3\} \cap \{D_1, \dots, D_k\} \neq \emptyset$.

Proof. By Theorem 8 a reversing g requires at least one component that is reversing at g , and by Table 1 the only components capable of reversing are $-4, \pm 8$ and -3 . The component -4 occurs iff $d \equiv 3 \pmod{4}$, a component ± 8 occurs iff d is even, and -3 occurs iff $3 \mid d$; this gives the stated equivalence of conditions. Conversely, if such a component D_j exists, choose g by CRT to lie in a reversing class of D_j and in the class 0 modulo $|D_i|$ for every $i \neq j$, which is possible as the moduli are coprime; then every other component preserves, exactly one reverses, and g may be taken even (if necessary replace g by $g + f$, noting f is even unless $f = d \equiv 1 \pmod{4}$ is odd, in which case no reversing component exists and the case does not arise). By Theorem 8, g is reversing. $\square$

Remark 10. The smallest bases with no exclusion law are therefore those with $d \in \{5, 13, 17, 29, 37, 41, 53, 61, \dots\}$, i.e. $d \equiv 1 \pmod{4}$, d odd, $3 \nmid d$. Among $2 \leq a \leq 30$ these are $a = 5, 13, 17, 20, 29$ (note $\text{sqf}(20) = 5$). At the other extreme, $d \equiv 3 \pmod{4}$ with $3 \mid d$ gives $f = 12$ and *three* of the six even classes modulo 12 are exclusion classes; this occurs for $a = 3, 12, 27, \dots$.

TABLE 2. Exclusion classes for selected bases. “Prime discs.” lists the factorization (4); the exclusion classes are those predicted by Theorem 8 and confirmed by exhaustive search.

a	f	prime discs.	exclusion classes mod f	preserving classes
2	8	8	4	0
3	12	$-3, -4$	4, 6, 8	0, 2, 10
5	5	5	<i>none</i>	<i>none</i>
6	24	$-3, -8$	8, 12, 16	0, 4, 20
7	28	$-7, -4$	14	0
10	40	5, 8	20	0
11	44	$-11, -4$	22	0
13	13	13	<i>none</i>	<i>none</i>
15	60	$-3, 5, -4$	20, 30, 40	0, 10, 50
17	17	17	<i>none</i>	<i>none</i>
21	21	$-3, -7$	14	<i>none</i>
29	29	29	<i>none</i>	<i>none</i>
30	120	$-3, 5, -8$	40, 60, 80	0, 20, 100

Computational confirmation of the classification. For each non-square a with $2 \leq a \leq 30$ we computed the reversing and preserving sets directly, by evaluating χ_a on every residue class in R_f attained by a prime below 2×10^5 and testing Definition 4 exhaustively over all even g modulo f . The resulting sets agree *exactly*, in every case, with the prediction of Theorem 8; and the criterion of Corollary 9 was confirmed for all non-square $2 \leq a \leq 79$. Table 2 records a representative sample.

3. DATA AND COMPUTATION

We enumerated the primes up to $x = 10^9$ by a segmented sieve of Eratosthenes with segment length 2^{22} , obtaining 50,847,532 primes $p \geq 5$ and hence 50,847,531 consecutive pairs. For each prime p we factored $p - 1$ once, by trial division over the primes below 10^5 together with the cofactor, and then tested primitive-root status for all of the bases

$$a \in \{2, 3, 5, 6, 7, 10, 11, 13, 17, 21, 29\}$$

simultaneously, using the standard criterion that a is a primitive root modulo p iff $a^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for every prime $\ell \mid p - 1$. Modular exponentiation was carried out in 128-bit arithmetic. Sharing the factorization of $p - 1$ across all eleven bases is what makes the computation inexpensive; the entire run takes under an hour on a single core and writes no intermediate table, accumulating instead the 2×2 contingency counts of (Artin status of p_n , Artin status of p_{n+1}), both globally and separately for each gap $g \leq 300$.

The base set was chosen to span the dichotomy of Corollary 9: $a = 5, 13, 17, 29$ admit no exclusion law, $a = 3$ and $a = 6$ have three exclusion classes each, and $a = 2, 7, 10, 11, 21$ have one. All counts reported below are exact; no sampling is involved. The implementation, together with an independent and slower verification in `sympy` used to cross-check the exclusion counts at 3×10^6 , is available at the repository cited in the data availability statement.

TABLE 3. Exclusion law at $x = 10^9$. “Pairs” counts consecutive prime pairs whose gap lies in an exclusion class for the given base; “both Artin” counts those in which a is a primitive root modulo both primes. The control column gives, for comparison, the proportion of doubly-Artin pairs at the preserving class $g \equiv 0$.

a	f	exclusion classes	pairs	both Artin	control (preserving)
2	8	4	13,404,106	0	27.5%
3	12	4, 6, 8	27,010,759	0	28.8%
6	24	8, 12, 16	12,419,039	0	29.3%
7	28	14	3,567,335	0	26.3%
10	40	20	2,214,511	0	26.7%
11	44	22	1,796,191	0	27.5%
21	21	14	2,693,804	0	—
<i>total</i>			63,105,745	0	

4. VERIFICATION OF THE EXCLUSION LAW AT 10^9

Theorem 5 is a proved statement, so a computation cannot strengthen it; but it is a sharp and easily falsified prediction about a large finite set, and checking it guards against errors in the classification of Theorem 8 and in the implementation. For each base with at least one exclusion class we counted the consecutive prime pairs (p_n, p_{n+1}) whose gap lies in an exclusion class, and among those the pairs that are doubly Artin. Table 3 reports the result.

There are 63,105,745 pairs in exclusion classes and not a single doubly-Artin pair among them, while at preserving classes the doubly-Artin proportion lies between 26.3% and 29.3% throughout, as one expects from the square of the relevant Artin density. (The control entry for base 21 is blank because base 21 has no preserving class at all, by Table 2.) To calibrate how striking the zero count is: absent the exclusion law, treating Artin statuses as independent with the observed per-base densities (≈ 0.374 throughout), the expected number of doubly-Artin pairs in the exclusion classes would be $\sum_a (\text{density}_a)^2 \times (\text{pairs}_a) \approx 8,821,000$ — some 1.9 million for base 2 alone and 3.8 million for base 3. The observed count is 0. Of course the theorem makes the zero certain, so this is a calibration of the *verification*, not of the theorem: an implementation or classification error affecting even a tiny fraction of pairs would have surfaced against an expected background of nearly nine million. A supplementary check reproduced the same zero counts, independently, with `sympy` order computations for primes below 3×10^6 .

5. THE CONSECUTIVE-PAIR CORRELATION ACROSS BASES

We now measure $\delta(a)$ of (1). Write n_{ij} for the number of consecutive pairs with Artin status i at p_n and j at p_{n+1} , so that

$$\hat{\delta}(a) = \frac{n_{11}}{n_{10} + n_{11}} - \frac{n_{01}}{n_{00} + n_{01}},$$

with the standard binomial standard error for a difference of two independent proportions. Table 4 collects the results.

A remark on the z -scores in Table 4 is in order. With fifty million pairs, even a minute dependence produces an enormous z ; the values up to -479 should not be read as hypothesis tests — at this sample size the null of exact independence was never seriously in doubt —

TABLE 4. Consecutive-pair Artin correlation at $x = 10^9$, over all 50,847,531 consecutive prime pairs, sorted by $|\delta|$. Here $\#exc$ is the number of exclusion classes and w the fraction of all consecutive pairs whose gap lies in one.

a	f	$\#exc$	w	$\widehat{\delta}(a)$	z
5	5	0	0.0000	-0.066563	-478.9
2	8	1	0.2636	-0.050199	-361.0
3	12	3	0.5312	-0.046676	-335.4
13	13	0	0.0000	-0.042573	-305.6
21	21	1	0.0530	-0.038345	-275.2
17	17	0	0.0000	-0.033025	-236.7
6	24	3	0.2442	-0.025205	-180.4
29	29	0	0.0000	-0.022422	-160.4
11	44	1	0.0353	-0.018909	-135.2
10	40	1	0.0436	-0.014140	-101.0
7	28	1	0.0702	-0.013491	-96.4

but simply as confirmation that sampling noise is negligible at the fourth decimal of δ . The scientifically relevant quantity throughout is the effect size $\delta(a)$ itself, not its significance.

Two features stand out. First, $\delta(a) < 0$ for every base, with $|z| \geq 96$: consecutive primes *repel* in Artin status, uniformly in the base. This is consistent with the mechanism proposed for the base-10 case, namely that the Lemke Oliver–Soundararajan bias [11] in the joint distribution of consecutive prime residues propagates through the factorization of $p - 1$ — and hence through $\omega(p - 1)$, whose consecutive values are themselves anticorrelated — into Artin status.

Second, and contrary to what the first half of this paper might suggest, the ordering of $|\delta(a)|$ has almost nothing to do with the exclusion law.

Observation 11 (Refutation of the exclusion-density hypothesis). The hypothesis that $|\delta(a)|$ increases with the density w of exclusion classes is false. At $x = 10^9$:

- (1) base 5, for which Corollary 9 shows no exclusion class exists, has the largest $|\delta|$ among the eleven bases;
- (2) base 3, whose exclusion classes contain 53% of all consecutive pairs, ranks third;
- (3) the mean of $|\delta|$ over the seven bases with an exclusion law is 0.0296, against 0.0411 over the four without, a ratio of 0.72 — opposite in direction to the hypothesis;
- (4) $r(w, |\delta|) = 0.23$ over the eleven bases, and the apparent weak positive association is confounded, since small conductors are exactly the even and 3-divisible ones.

The reason is one of scale rather than of mechanism, and in hindsight it is clear. An exclusion class is an absolute prohibition, but it applies only to certain gap classes; the statistic $\delta(a)$ averages over all consecutive pairs, in which those classes are a minority, so the prohibition is diluted. Indeed the exclusion classes contribute to δ only through the pairs they contain, and their effect is bounded by w times a quantity of order the Artin density — for base 10, where $w = 0.044$, this bound is already smaller than the observed $|\delta|$.

What does organise the data is the conductor.

Observation 12 (Conductor law). Over the eleven bases at $x = 10^9$,

$$r(\log f, |\delta|) = -0.957, \quad r(f^{-1/2}, |\delta|) = +0.951, \quad r(f^{-1}, |\delta|) = +0.921,$$

against $r(w, |\delta|) = +0.231$ for the exclusion density. Thus $|\delta(a)|$ is a decreasing function of the conductor of χ_a , and this single parameter accounts for the observed ordering far better than any exclusion-based quantity. Since a correlation computed from eleven points can flatter, we also report a nonparametric bootstrap (10^5 resamples of the eleven bases): the 95% interval for $r(\log f, |\delta|)$ is $[-0.99, -0.86]$, bounded well away from zero, though the reader should bear in mind that the eleven conductors are not a random sample from any population.

The heuristic is that χ_a is determined modulo f , so the residue information relevant to Artin status base a is carried by the $\varphi(f)$ classes in $(\mathbb{Z}/f\mathbb{Z})^\times$. The transition bias of [11] is distributed across these channels; the fewer the channels, the less the cancellation between them, and the larger the surviving global correlation. Figure 1 displays both associations.

We stress that Observation 12 is at present a purely empirical regularity, and we indicate what a derivation would involve. Write the consecutive-pair correlation as a sum over joint residue channels,

$$\delta(a) \approx \sum_{r,s \in (\mathbb{Z}/f\mathbb{Z})^\times} \left(\pi(r,s) - \pi(r)\pi(s) \right) \alpha_a(r) \alpha_a(s),$$

where $\pi(r,s)$ is the density of consecutive prime pairs with $p_n \equiv r, p_{n+1} \equiv s \pmod{f}$, and $\alpha_a(r)$ is the deviation of the Artin density in the class r from its mean. The first factor is governed, conjecturally, by the Hardy–Littlewood singular series [6] through the conjectural formula of Lemke Oliver–Soundararajan [11, Conj. 1.1] for prime transitions in residue classes; the second by Hooley’s conditional density computation [8] restricted to residue classes, in which $\chi_a(r) = -1$ is the leading constraint. Carrying this out — in particular extracting the f -dependence of the double sum, where the number of channels $\varphi(f)^2$ competes against the per-channel bias — would replace the measured correlations above with a formula and settle the exponent question. We have not done this here: the LOS transition term for *consecutive* (rather than merely distinct) primes in fixed residue classes involves a secondary term with no closed form, and the interaction with the $\omega(p-1)$ channel (through which the bias reaches Artin status) is not multiplicative. The calculation appears feasible but is a paper of its own, and we leave it as the first item of future work.

We deliberately refrain from asserting an exact power law. The products $|\delta| \cdot f$ drift from 0.33 to 0.83 across the range, while $|\delta| \cdot \sqrt{f}$ drift only from 0.15 to 0.13; so $f^{-1/2}$ is the better one-parameter description, but neither is constant, and with eleven bases spanning $5 \leq f \leq 44$ we regard the evidence as insufficient to identify an exponent. The honest statement is Observation 12: a strong monotone dependence on f , whose precise shape should be derived from, rather than fitted ahead of, a channel-by-channel analysis.

Remark 13. The two smallest $|\delta|$ belong to $a = 7$ and $a = 10$, with $f = 28$ and $f = 40$; the largest to $a = 5$ with $f = 5$. Note that $a = 21$, with $f = 21$, sits above both $a = 17$ ($f = 17$) and $a = 6$ ($f = 24$), so the dependence on f is not perfectly monotone — as one would expect if the relevant quantity is a sum over the $\varphi(f)$ channels rather than a function of f alone.

6. DISCUSSION

What is new here. The ingredients are classical: the quadratic obstruction to being a primitive root, quadratic reciprocity, the decomposition of a discriminant into prime discriminants, and the elementary evaluation of a Jacobi-type character sum. The contributions are (i) the observation that these combine into a *deterministic exclusion law* for consecutive Artin primes, valid in every base (Theorem 5); (ii) a *complete classification* of the exclusion classes in terms of prime discriminants (Theorem 8), with the resulting dichotomy (Corollary 9) identifying exactly

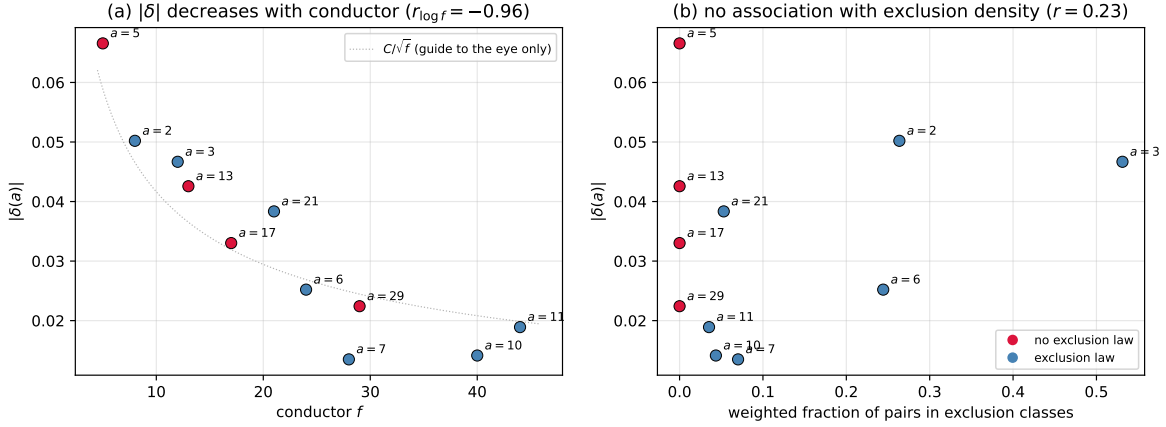


FIGURE 1. Left: $|\delta(a)|$ against the conductor f of χ_a at $x = 10^9$. The data points are the content of the figure; the faint dotted curve $C/\sqrt{f}$ ($C = 0.132$) is a guide to the eye only — as discussed after Observation 12, the data do not identify an exponent and no power law is asserted. Right: $|\delta(a)|$ against the weighted fraction of consecutive pairs lying in exclusion classes. Bases admitting no exclusion law are shown in red; one of them has the largest $|\delta|$ of all.

which bases admit no such law; and (iii) the first cross-base measurement of the consecutive-pair Artin correlation, which refutes the natural hypothesis that the exclusion law drives that correlation and identifies conductor size as the operative parameter instead.

We regard (iii) as no less useful than (i) and (ii). The exclusion law is visually striking — tens of millions of pairs with an exact zero count — and it would be easy to assume that it dominates the statistics of Artin primes in pairs. It does not. Separating a locally absolute but globally negligible constraint from the globally operative conductor effect seems to us the right conclusion to draw, and it is one we reached only by measuring.

Relation to earlier work. For base 10 the exclusion class $g \equiv 20 \pmod{40}$ and a measurement of $\delta(10)$ were obtained by the author in earlier work; the present paper subsumes that case, since $d = 10$ gives $D = 40 = 5 \cdot 8$ and Theorem 8 returns the single exclusion class $g \equiv 20$. Garcia–Kahoro–Luca [3] study primitive-root bias for twin primes, i.e. the single gap $g = 2$; our Table 1 shows that $g = 2$ is an exclusion class exactly when some reversing component has $g = 2$ in its reversing class and no component is indeterminate there, which for $g = 2$ requires $f \mid 12$ — so $a \equiv 3, 12, 27, \dots$ are the twin-prime cases covered by our law. Pollack [14] and Baker–Pollack [1] construct, under GRH, bounded-gap runs of primes sharing a primitive root; our results add a base-specific constraint on where such runs can occur, since a run in base a must avoid every exclusion class of a , and for $a = 3$ this excludes half of the available even gaps.

Future work. Three directions suggest themselves. First, the channel decomposition displayed after Observation 12 should be carried out: a derivation from the singular series of [6] together with Hooley’s density computation [8], summed over the $\varphi(f)$ residue channels, would replace an empirical regularity with a formula and settle the exponent question. Second, the decay of $\delta(a)$ in x should be measured for several bases at once; for base 10 a decay of order $1/\log x$ was conjectured, and the present pipeline extends to 10^{12} without difficulty. Third, the classification invites an analogue for higher-order obstructions: membership in A_a also requires a to be a non- ℓ -th-power residue for every $\ell \mid p - 1$, and the cubic case in particular should produce

exclusion laws governed by conductors of cubic characters, with $\mathbb{Q}(\sqrt{-3})$ playing a distinguished role.

ACKNOWLEDGEMENTS

The author thanks the developers of the open-source tools used in this work (GCC, Python, `sympy`, `matplotlib`).

Use of generative AI. The author used Anthropic’s Claude (models Claude Opus 4.5 and Claude Opus 5) as an assistant in this work: for drafting and revising exposition, for writing and debugging the sieve and analysis programs, and for exploratory discussion of the classification. Two specific interventions should be recorded in the interest of transparency. First, an initial machine-generated form of Table 1 incorrectly classified the conductor-8 component as preserving at $g \equiv 2 \pmod{8}$; the error was detected by exhaustive computational comparison against Theorem 8 and corrected before any proof was written. Second, the hypothesis refuted in Observation 11 was originally proposed, with a plausible heuristic, in the course of that assisted exploration; it was tested and rejected on the evidence. All mathematical statements in the final paper were verified by the author, and every theorem is accompanied by a proof that does not depend on machine assistance. The author is solely responsible for the content.

DECLARATION OF INTEREST

The author reports there are no competing interests to declare.

FUNDING

This research received no external funding.

DATA AVAILABILITY

All code and results are publicly available at <https://github.com/jpbald93/consecutive-artin>, including the segmented sieve and multi-base correlation program, the classification and verification scripts, the exact contingency tables underlying Tables 3 and 4, and scripts reproducing Figure 1. The computation reported here can be reproduced from the repository in under an hour on a single core.

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